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Sydney Girls High School



2000 Yearly Exam

MATHEMATICS 2U/3U

Year 11

Time allowed - 90 minutes

Instructions

NAME _____

- Attempt all eight questions.
- Questions are NOT of equal value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Each question attempted should be started on a new sheet. Write on one side of the paper only.

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Question One (10 marks)

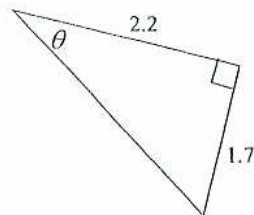
- Evaluate $\frac{1}{\sqrt{5.3^2 - 1.7}}$ to 2 decimal places
- Evaluate $|-5| - |7|$
- Evaluate $8.3 \times 10^{15} - 7.1 \times 10^{13}$ and give your answer in scientific notation correct to 2 significant figures.
- Simplify $2 - 2(2x - 1)$
- Write $2\sqrt{44}$ in simplest surd form.
- Remove the grouping symbols: $(\sqrt{3} + 4)(\sqrt{2} - \sqrt{5})$
- Solve for x and plot your solution on a number line: $-5 < 2x + 7 \leq 11$

Question Two (10 marks)

- Write $\sqrt[3]{(3+x)^2}$ in index form
- Simplify $2ab - b^2 + 5ab - 2b^2$
- Factorise $4y^2 - 1$
- Factorise $x^2 - 5x + 2xy - 10y$
- Factorise $25 - (x - b)^2$
- If $\tan \theta = \frac{6}{11}$ and θ acute, find the exact ratios of $\sin \theta$ and $\cos \theta$.

Question Three (12 marks)

- a) Solve $2x^2 + 3x - 7 = 0$ (leave your answer in exact form)
 b) Find θ in the triangle below;



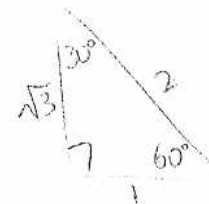
- c) A plane leaves Sydney and flies on a bearing of 235 for 750 km.
 i) Draw a diagram and mark this information on it.
 ii) Calculate how far west of Sydney the plane is.
 d) i) Sketch $x^2 + y^2 = 9$
 ii) Is this graph a function? (give a reason for your answer)
 e) Solve $|2x + 1| = 5$

Question Four (13 marks)

- a) Simplify $\frac{a-3}{a^3-27}$
 b) Sketch $y = x^2 - 4$ on a number plane, and state its domain and range.
 c) Sketch the graph of $y = \cos x$ for $0 \leq \theta \leq 360$.
 d) A straight line has x -intercept 1 and y -intercept -2. Find its equation.
 e) Simplify $\sqrt{9 - 9\cos^2 \alpha}$

Question Five (12 marks)

- a) Solve $\frac{x}{2} - \frac{x+1}{5} = 1$
 b) If $A = \frac{1}{2}h(a+b)$, find the value of a if $A = 100$, $h = 10$ and $b = 8$.
 c) For the function $F(x) = x^4 + 1$;
 i) Find $F(1)$
 ii) Find x if $F(x) = 17$
 iii) Show that $F(x)$ is an even function
 d) Solve $2\sin\theta = 1$ for $0 \leq \theta \leq 360$.



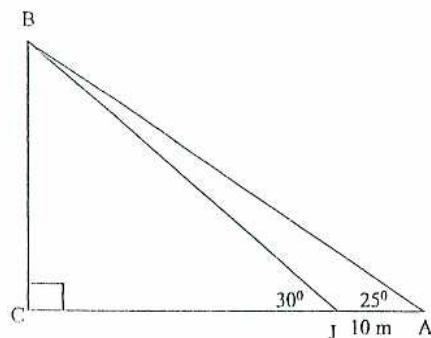
Question Six (16 marks)

Draw sketch graphs of each function showing relevant features. (do not use calculus)

- | | |
|-------------------------|--|
| a) $y = 2$ | b) $x + y = 2$ |
| c) $y = -2^x$ | d) $y = \frac{1}{x+2}$ |
| f) $y = 2+x $ | g) $y = x^2 - 2 $ |
| h) $y = \frac{-2}{x^2}$ | i) $f(x) = \begin{cases} 2x-1 & \text{if } x > 2 \\ x^2 & \text{if } x \leq 2 \end{cases}$ |

Question Seven (14 marks)

- a) Angela measures the angle of elevation of the top of a tower as 25° and Jessica measures the angle of elevation as 30° . Jessica is standing 10 metres closer to the tower than Angela



- i) Copy the diagram onto your test paper
 - ii) Show that $BJ = \frac{10 \sin 25^\circ}{\sin 5^\circ}$
 - iii) Hence find the height of the tower.
- b) Plot points A (2,1) and B (-2,-3) on a number plane.
- i) Find the midpoint, C, of AB.
 - ii) Show that the line through C perpendicular to AB has equation $x + y + 1 = 0$
 - iii) Show that this line passes through D(-3,2).
 - iv) Find the area of triangle ABD

Question Eight(13 marks)

- a) Find the exact value of $\cos 315^\circ$.
- b) Find the equation of the straight line passing through (-3,-2) and passing through the intersection of $2x - 5y - 3 = 0$ and $3x - 4y - 8 = 0$.
- c) Solve $2^{3x-1} = \frac{1}{4}$
- d) Given $x = \sqrt{5} + 2$, find b if $x + \frac{1}{x} = 2\sqrt{b}$
- e) Prove $(\sec^2 \theta)(\cos^2 \theta) = \sec^2 \theta + \cos^2 \theta$

$\alpha\beta\chi\delta\epsilon\phi\eta\theta\iota\kappa\lambda\mu\nu\rho\sigma\tau\upsilon\omega\xi\psi\zeta$

Q1.

a) 0.19

b) $5 - 7 = -2$

c) $8 \cdot 2 \times 10^{15}$

d) $2 - 4x + 2$
 $= 4 - 4x$

e) $2\sqrt{44} = 2 \times 2\sqrt{11}$
 $= 4\sqrt{11}$

f) $(\sqrt{3} + 4)(\sqrt{2} - \sqrt{5})$
 $= \sqrt{6} - \sqrt{15} + 4\sqrt{2} - 4\sqrt{5}$

g) $-5 < 2x + 7 \leq 11$
 $-12 < 2x \leq 4$
 $-6 < x \leq 2$



Q2

a) $\sqrt[3]{(3+x)^4} = (3+x)^{\frac{4}{3}}$

b) $7ab - 3b^2$

c) $4y^2 - 1 = (2y+1)(2y-1)$

d) $x^2 - 5x + 2xy - 10y$
 $x(x-5) + 2y(x-5)$
 $(x+2y)(x-5)$

e) $25 - (x-b)^2$
 $= [5 - (x-b)][5 + (x-b)]$
 $= (5-x+b)(5+x-b)$

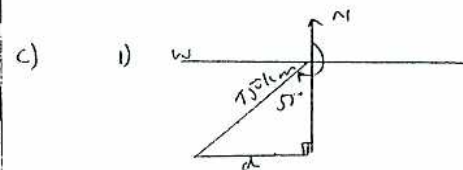
f)
 $\cos \theta < 0$ as θ in Q_2, Q_3
 $\sin \theta = \frac{6}{\sqrt{157}}$, $\tan \theta = \frac{6}{11}$
 $\cos \theta = \frac{11}{\sqrt{157}}$

Q3

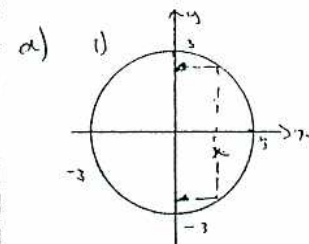
a) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 $= \frac{-3 \pm \sqrt{9 - 4(2)(-7)}}{4}$
 $= \frac{-3 \pm \sqrt{65}}{4}$

b) $\tan \theta = \frac{1.7}{2.2}$

$\theta = 55^\circ$



$\frac{d}{150} = \sin 55^\circ$
 $d = 150 \sin 55^\circ$
 $= 124 \text{ km}$



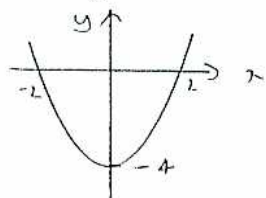
ii) No as for same given x there are 2 y 's (see diagram)

e) $2x+1 = 5$ or $2x+1 = -5$
 $2x = 4$ $2x = -6$
 $x = 2$ $x = -3$

Q4

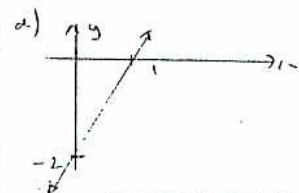
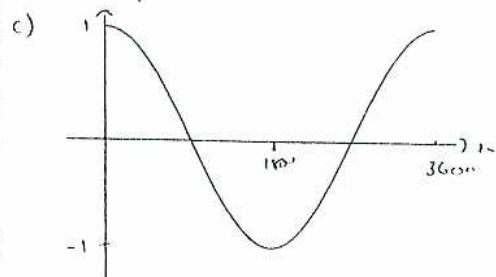
$$a) \frac{a-3}{a^2-27} = \frac{a-3}{(a-3)(a^2+3a+9)} = \frac{1}{a^2+3a+9}$$

b) $y = x^2 - 4$



x : all real

$$y \geq -4$$



$$y = 2x - 2$$

e)

$$\begin{aligned} \sqrt{9 - 9 \cos^2 \theta} &= \sqrt{9(1 - \cos^2 \theta)} \\ &= \sqrt{9 \sin^2 \theta} \\ &= 3 \sin \theta \end{aligned}$$

Q5

a)

$$\begin{aligned} \frac{x}{2} - \frac{x+1}{5} &= 1 \\ 5x - 2(x+1) &= 10 \\ 5x - 2x - 2 &= 10 \\ 3x &= 12 \\ x &= 4 \end{aligned}$$

b)

$$\begin{aligned} A &= \frac{1}{2}h(a+b) \\ 100 &= \frac{1}{2} \cdot 5(a+8) \\ 20 &= a+8 \\ a &= 12 \end{aligned}$$

c) i) $F(x) = x^4 + 1$
 $= 2$

ii) $17 = x^4 + 1$
 $16 = x^4$
 $x = \pm 2$

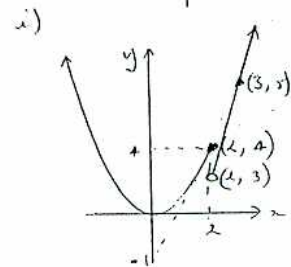
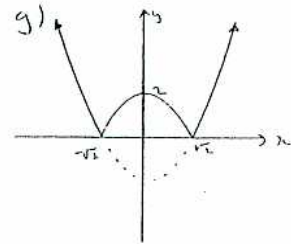
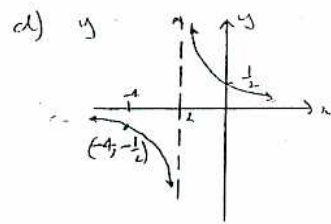
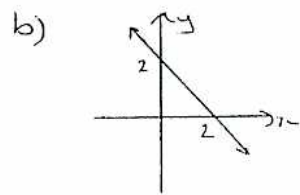
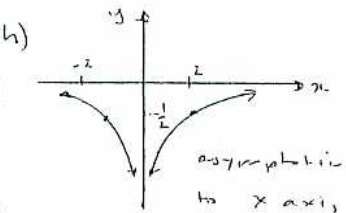
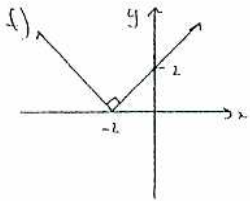
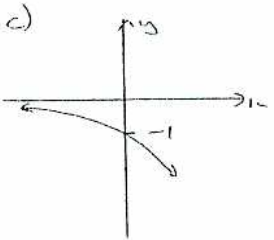
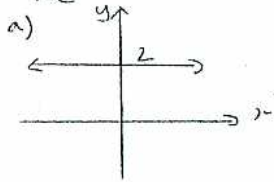
iii) Even if $F(x) = F(-x)$
 $F(x) = x^4 + 1$, $F(-x) = (-x)^4 + 1 = x^4 + 1$

$\therefore$ Even

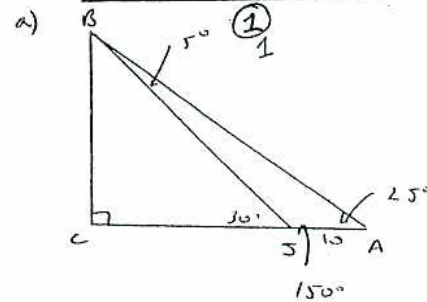
d)

$$\begin{aligned} 2 \sin \theta &= 1 \\ \sin \theta &= \frac{1}{2} \\ \theta &= 30^\circ, 150^\circ \end{aligned}$$

Q6



Question Seven



ii) $\frac{BS}{\sin 25^\circ} = \frac{10}{\sin 5^\circ}$

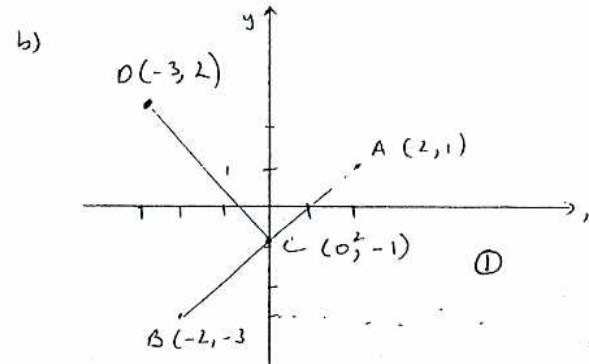
$$BS = \frac{10 \sin 25^\circ}{\sin 5^\circ} \quad ①$$

iii) $\frac{BC}{BS} = \sin 30^\circ$

$$BC = BS \times \sin 30^\circ$$

$$= \frac{10 \sin 25^\circ \sin 30^\circ}{\sin 5^\circ}$$

$$BC = 24.2 \text{ m} \quad ②$$



i) $\cos \theta = \frac{x_2 - x_1}{2} = 0$

$$x = \frac{2 + (-2)}{2} = 0 \quad ②$$

$$y = \frac{1 - (-3)}{2} = -1$$

ii) gradient AB

$$m_1 = \frac{1 - (-3)}{2 - (-2)} = 1 \quad ①$$

$\therefore m_2 = -1$ ($m_1, m_2 = -1$)

Eqn $y - y_1 = m(x - x_1)$

$$y + 1 = -1(x - 0)$$

$$y + 1 = -x$$

$$x + y + 1 = 0 \quad ①$$

iii) substit $x = -3, y = 2$ in above

$$-3 + 2 + 1 = 0 \quad ①$$

iv) $AB = \sqrt{4^2 + 4^2} = 4\sqrt{2} \quad ①$

$CO = \sqrt{3^2 + 3^2} = 3\sqrt{2} \quad ①$

$A = \frac{1}{2} \times \text{base} \times \text{height}$

$$= \frac{1}{2} \times 4\sqrt{2} \times 3\sqrt{2}$$

$$= 12 \text{ units}^2 \quad ①$$

Question 8

$$a) \cos 315^\circ = \cos 45^\circ \\ = \frac{1}{\sqrt{2}}$$

$$b) \text{ Eqn of form } 2x - 5y - 3 + k(3x - 4y - 8) = 0$$

$$x = -3, y = -2$$

$$-6 + 10 - 3 + k(-9 + 8 - 8) = 0$$

$$1 - 9k = 0$$

$$k = \frac{1}{9}$$

$$\text{ie } 2x - 5y - 3 + \frac{1}{9}(3x - 4y - 8) = 0$$

$$18x - 45y - 27 + 3x - 4y - 8 = 0$$

$$21x - 49y - 35 = 0$$

$$c) 2^{3x-1} = \frac{1}{8}$$

$$3 \cdot 2^{3x-1} = 2^{-2}$$

$$3x - 1 = -2$$

$$3x = -1$$

$$x = -\frac{1}{3}$$

$$d) \text{ If } x + \frac{1}{x} = \sqrt{5} + 2 \quad \text{then}$$

$$x + \frac{1}{x} = \sqrt{5} + 2 + \frac{1}{\sqrt{5} + 2}$$

$$= \sqrt{5} + 2 + \sqrt{5} - 2$$

$$= 2\sqrt{5} \quad \text{ie } b = 5$$

$$e) \text{ Prove } (\sec^2 \theta)(\operatorname{cosec}^2 \theta) = \sec^2 \theta + \operatorname{cosec}^2 \theta$$

$$\text{RHS} = \sec^2 \theta + \operatorname{cosec}^2 \theta$$

$$= \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{1}{\sin^2 \theta \cos^2 \theta}$$

$$= \sec^2 \theta \operatorname{cosec}^2 \theta$$